



Reconstruction-Based Anomaly Detection on ECG5000:

A Comparative Study of Conv1D and LSTM Autoencoders with
Classical Baselines

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Abstract

We study reconstruction-based anomaly detection on the **ECG5000** heartbeat time-series dataset. A 1D convolutional autoencoder (Conv1D AE) and an LSTM autoencoder (LSTM AE) are trained on *normal heartbeats only* under an identical experimental protocol and compared against two classical baselines: One-Class SVM (OC-SVM) and Isolation Forest. We systematically compare four threshold-selection strategies on validation data and freeze the final threshold before evaluating on the untouched test set of 4500 heartbeats. On this test set the Conv1D AE reaches $F1=0.943$, $ROC-AUC=0.988$ and $PR-AUC=0.967$, outperforming the LSTM AE ($F1=0.912$) and Isolation Forest ($F1=0.538$) while remaining competitive with OC-SVM ($F1=0.961$). A latent-dimension capacity ablation over $\{4, 8, 16, 32, 64\}$ exhibits a non-monotonic relationship between capacity and detection quality, in line with the well-known *reconstruction paradox*. Error analysis shows that false negatives concentrate on abnormal class 4 (supra-ventricular premature / ectopic beats), whose morphology is closest to the normal distribution. All results are produced by a single reproducible Jupyter notebook shipped with the report.

Contents

1	Introduction	3
2	Problem statement	3
3	Dataset	3
4	Data preprocessing	4
5	Problem formulation	5
6	Methodology	5
6.1	Conv1D autoencoder	5
6.2	LSTM autoencoder	6
6.3	Training hyperparameters	6
6.4	Threshold-selection strategies	7
6.5	Classical baselines	8
7	Experimental setup	8
8	Results	8
8.1	Threshold-selection comparison (Conv1D)	8
8.2	Conv1D vs. LSTM	9
8.3	Classical baseline comparison	10
8.4	Failure-case analysis	11
8.5	Capacity ablation	12
9	The reconstruction paradox	13
10	Self-attention discussion	13
11	Limitations	14
12	Conclusion	14

1 Introduction

Anomaly detection on physiological time series is a canonical application of unsupervised and semi-supervised data-mining techniques [1, 12]. ECG5000, a subset of a 20-hour Holter recording curated by Chen *et al.* [2] and distributed through the UCR time-series archive [3, 5], is a widely used benchmark for time-series anomaly detection because the majority of its 5 000 heartbeats are normal while abnormal beats form several heterogeneous minority classes.

This project pursues four goals:

- (i) build a leakage-free reconstruction-based pipeline;
- (ii) compare a **Conv1D autoencoder** against an **LSTM autoencoder** [6, 7, 11];
- (iii) benchmark them against classical anomaly detectors, OC-SVM [14] and Isolation Forest [10];
- (iv) study threshold-selection and model-capacity sensitivity.

2 Problem statement

Given a heartbeat $x \in \mathbb{R}^T$ with $T = 140$, we build a scoring function $s : \mathbb{R}^T \rightarrow \mathbb{R}_{\geq 0}$ and a threshold τ such that

$$\hat{y}(x) = \mathbf{1}[s(x) > \tau], \quad \hat{y}(x) \in \{0, 1\}$$

denotes normal vs. abnormal. The scoring function must be learned without ever seeing abnormal beats during training of the autoencoder (semi-supervised anomaly detection).

3 Dataset

ECG5000 provides an official train/test split of 500/4 500 heartbeats of length $T=140$. There is one integer label per heartbeat with five possible values. Class 1 (*Normal*) is the majority class; classes 2–5 correspond to different abnormal beats. Table 1 summarises the dataset and Table 2 the class distribution.

Table 1: Dataset summary.

Property	Value
Total heartbeats	5 000
Time steps per heartbeat (T)	140
Official training rows	500
Official test rows	4 500
Number of classes	5
Missing values (train / test)	0 / 0
Duplicate rows (train / test)	0 / 0
Amplitude range (train)	$[-5.80, 4.06]$

Table 2: Class distribution.

Class	Train	Test	Description
1	292	2 627	Normal beat (N)
2	177	1 590	R-on-T PVC
3	10	86	Premature ventricular contraction (PVC)
4	19	175	Supra-ventricular premature or ectopic beat (SP/EB)
5	2	22	Unclassifiable beat (UB)
Total	500	4 500	

For anomaly detection the labels are collapsed into a binary target: $y = 0$ (Normal, original class 1) and $y = 1$ (Abnormal, classes 2–5). This yields 292 normal and 208 abnormal in the training portion and 2 627 normal and 1 873 abnormal in the test portion.

Figure 1 displays one representative example per class. Figure 2 shows the class distribution.

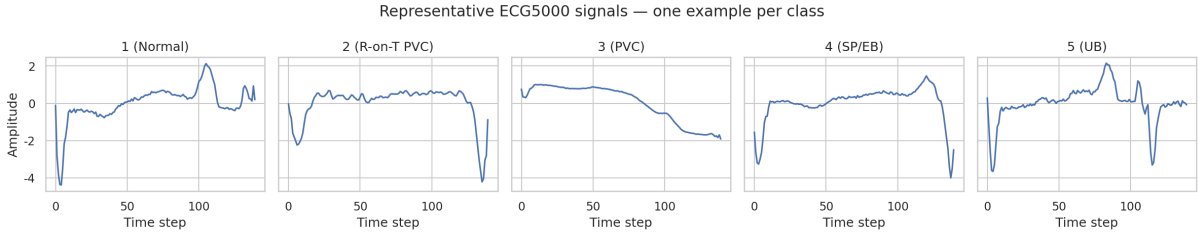


Figure 1: Representative ECG heartbeats — one example per class.

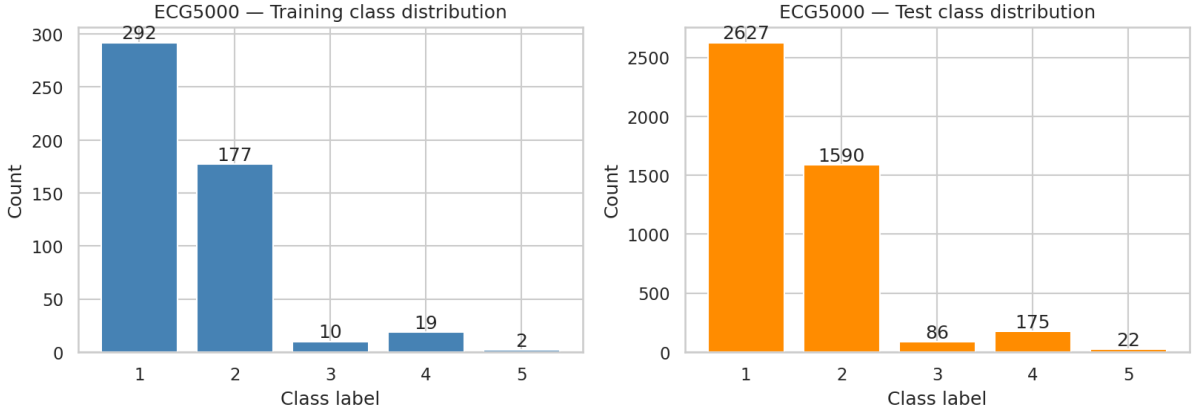


Figure 2: Class distribution in the training and test partitions.

4 Data preprocessing

The training set is further split into 400 training rows and 100 validation rows, stratified on the binary label with random seed 42. From the training portion we keep only the 234 normal rows for autoencoder training. Standardisation is fit on those 234 rows:

$$x_{\text{scaled}} = \frac{x - \mu_{\text{train,normal}}}{\sigma_{\text{train,normal}}}, \quad \mu \approx 0, \sigma \approx 0.9964,$$

and applied unchanged to validation and test. The empirical mean is already extremely close to zero because each heartbeat in the UCR distribution has been amplitude-normalised. Table 3 lists the preprocessing configuration.

Table 3: Preprocessing configuration.

Item	Value
Split	Official (500 train / 4 500 test)
Validation carved from	Training portion (20%, stratified, seed 42)
Scaler	Global z -score
Scaler fit on	Normal training rows only ($n=234$)
Applied to	Train, validation, test
$\mu_{\text{train,normal}}$	≈ 0
$\sigma_{\text{train,normal}}$	0.9964

5 Problem formulation

An autoencoder $f_\theta : \mathbb{R}^T \rightarrow \mathbb{R}^T$ is a composition $f_\theta = g_\phi \circ h_\psi$ of a decoder g_ϕ and an encoder h_ψ . Training minimises the empirical reconstruction risk on normal training data only:

$$\theta^* = \arg \min_{\theta} \frac{1}{N_n} \sum_{i:y_i=0} \|x_i - f_\theta(x_i)\|_2^2.$$

The anomaly score is the per-sample MSE:

$$s_i = \frac{1}{T} \sum_{t=1}^T (x_{i,t} - \hat{x}_{i,t})^2.$$

An anomaly is declared whenever $s_i > \tau$.

MSE is used both as training loss and as anomaly score. This coherence guarantees that the objective minimised during training is exactly the quantity that later drives detection: it removes the risk of a train-metric / evaluation-metric mismatch. Its symmetry over the $T=140$ time steps also makes it insensitive to the location of a morphological deviation within the beat.

6 Methodology

6.1 Conv1D autoencoder

The Conv1D AE uses three convolutional stages with ReLU non-linearities, symmetric max-pool downsampling and linear upsampling, and a fully-connected bottleneck. The architecture is summarised in Table 4.

Table 4: Conv1D autoencoder architecture. All Conv1D layers use stride 1 and SAME-style padding.

Stage	Layer	Output shape	Trainable?
Input	–	1×140	–
Enc. 1	Conv1D $1 \rightarrow 16$, $k = 7$ + ReLU	16×140	yes
	MaxPool /2 + Dropout(0.10)	16×70	no
Enc. 2	Conv1D $16 \rightarrow 32$, $k = 5$ + ReLU	32×70	yes
	MaxPool /2 + Dropout(0.10)	32×35	no
Enc. 3	Conv1D $32 \rightarrow 64$, $k = 3$ + ReLU	64×35	yes
Bottleneck	Flatten + Linear	16	yes
Dec. 0	Linear + reshape	64×35	yes
Dec. 1	Conv1D $64 \rightarrow 32$, $k = 3$ + ReLU + Up $\times 2$	32×70	yes
Dec. 2	Conv1D $32 \rightarrow 16$, $k = 5$ + ReLU + Up $\times 2$	16×140	yes
Output	Conv1D $16 \rightarrow 1$, $k = 7$	1×140	yes
Total parameters			91,729

6.2 LSTM autoencoder

The LSTM AE operates on the temporal dimension. The 140-step signal is reshaped to $(B, T, 1)$, passed through a single-layer LSTM with 32 hidden units, and the last hidden state is projected to a 16-dimensional latent vector. A decoder LSTM then unrolls the same latent vector for 140 steps and a linear head produces the reconstruction. Table 5 summarises the architecture.

Table 5: LSTM autoencoder architecture. Hidden size 32, latent dim 16.

Stage	Layer	Output shape
Input	$(B, 1, 140) \rightarrow (B, 140, 1)$	–
Encoder LSTM	1 layer, hidden 32	$(B, 32)$ (final h)
Enc-to-latent linear	$32 \rightarrow 16$ + Dropout	$(B, 16)$
Latent-to-dec linear	$16 \rightarrow 32$, repeat $T = 140$	$(B, 140, 32)$
Decoder LSTM	1 layer, hidden 32	$(B, 140, 32)$
Output linear	$32 \rightarrow 1$ + Dropout	$(B, 140, 1)$
Total parameters		14,033

6.3 Training hyperparameters

Both networks are optimised with Adam [9] and MSE loss under matched protocol (Table 6). Early stopping monitors the reconstruction MSE on the *normal* validation subset.

Table 6: Training hyperparameters (shared by Conv1D and LSTM AE).

Hyperparameter	Value
Optimiser	Adam ($\beta_1=0.9$, $\beta_2=0.999$)
Learning rate	10^{-3}
Weight decay	10^{-5}
Batch size	32
Maximum epochs	100
Early-stopping patience	12 epochs
Latent dimension	16
Dropout	0.10
Reconstruction loss	MSE (mean over $T=140$)
Seed (Python, NumPy, PyTorch)	42

Training curves for the Conv1D AE are shown in Figure 3.

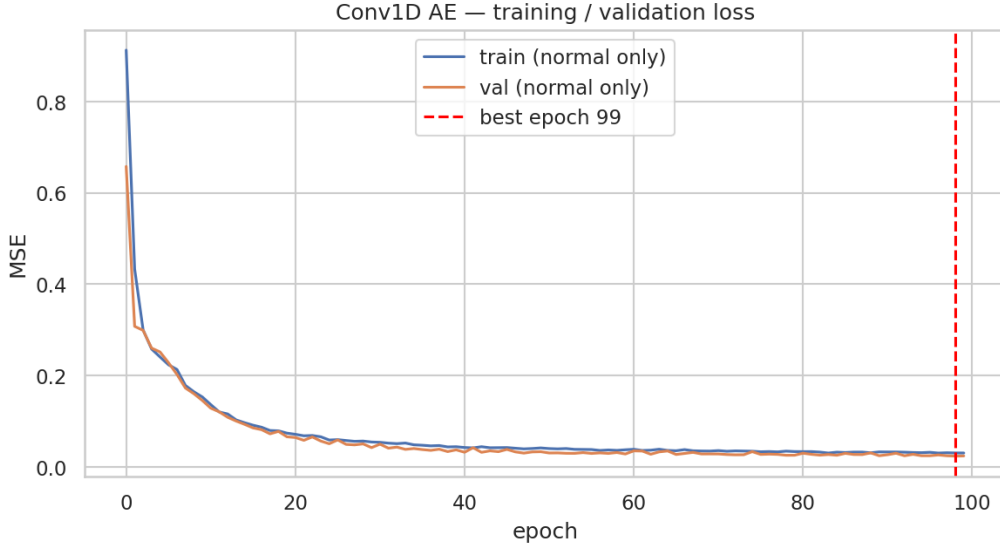


Figure 3: Conv1D AE training / validation MSE on normal data only. Best validation epoch: 99.

6.4 Threshold-selection strategies

Four strategies are compared on the validation set:

- **Percentile:** $\tau = P_p(\{s_i : y_i = 0\}_{\text{val}})$ for $p \in \{90, 92.5, 95, 97.5, 99\}$.
- **Mean+ $k\sigma$:** $\tau = \mu_n + k\sigma_n$ with μ_n, σ_n computed on normal validation errors and $k \in \{1, 1.5, 2, 2.5, 3\}$.
- **Median+ k MAD:** $\tau = \text{med}(s) + k \cdot \text{MAD}(s)$ with $\text{MAD}(s) = \text{med}(|s - \text{med}(s)|)$ and $k \in \{1, \dots, 5\}$.
- **Validation-F1 argmax:** the τ^* maximising F1 on the (labelled) validation set.

The validation-F1 strategy is legitimate because the validation set is expressly reserved for model / threshold selection; it is *never* applied to the test set. Once τ^* is chosen it is frozen and the test set is scored exactly once.

6.5 Classical baselines

One-Class SVM [14] is trained on the same normal training pool with an RBF kernel, $\gamma = \text{scale}$, $\nu = 0.05$; anomaly score is the negative of the decision function. **Isolation Forest** [10] is trained on the full 400-row training set with 200 estimators; contamination is set to the empirical training abnormal fraction ($208/500 \approx 0.42$). The anomaly score is the negative decision function.

7 Experimental setup

Every result reported below is produced by a single reproducible notebook run on CPU under PyTorch 2.13. We report Precision, Recall, F1, ROC-AUC, PR-AUC and the full confusion matrix on the untouched test set. Accuracy is not the primary metric: with a $\approx 42\%$ abnormal test prevalence a constant classifier already reaches $\approx 58\%$ accuracy without detecting any anomaly. Precision-Recall AUC is particularly informative under class imbalance [13, 4].

8 Results

8.1 Threshold-selection comparison (Conv1D)

Table 7 compares the four strategies on validation data. The validation-F1-optimal threshold reaches $F1 = 0.963$ and is therefore selected as the final threshold for the Conv1D AE, $\tau^* = 0.11153$.

Table 7: Threshold-selection comparison on the validation set (Conv1D AE).

Method	Parameter	Threshold	Precision	Recall	F1
Percentile	$p = 90$	0.0351	0.875	1.000	0.933
Percentile	$p = 92.5$	0.0428	0.894	1.000	0.944
Percentile	$p = 95$	0.0583	0.932	0.976	0.953
Percentile	$p = 97.5$	0.0907	0.952	0.952	0.952
Percentile	$p = 99$	0.0984	0.976	0.952	0.964
Mean+ $k\sigma$	$k = 1.0$	0.0429	0.894	1.000	0.944
Mean+ $k\sigma$	$k = 1.5$	0.0524	0.913	1.000	0.955
Mean+ $k\sigma$	$k = 2.0$	0.0618	0.932	0.976	0.953
Mean+ $k\sigma$	$k = 2.5$	0.0713	0.930	0.952	0.941
Mean+ $k\sigma$	$k = 3.0$	0.0807	0.930	0.952	0.941
Median+ $k\text{MAD}$	$k = 1$	0.0230	0.689	1.000	0.816
Median+ $k\text{MAD}$	$k = 2$	0.0281	0.764	1.000	0.866
Median+ $k\text{MAD}$	$k = 3$	0.0333	0.857	1.000	0.923
Median+ $k\text{MAD}$	$k = 4$	0.0385	0.894	1.000	0.944
Median+ $k\text{MAD}$	$k = 5$	0.0436	0.894	1.000	0.944
Val-F1 argmax	—	0.1115	1.000	0.929	0.963

Threshold sensitivity on the test error distribution is shown in Figure 5: F1 varies smoothly across a wide range, implying that the selection is not brittle. Figure 4 shows the corresponding test-set reconstruction-error distributions: normal and abnormal errors are well separated, with the frozen threshold cleanly slicing the two populations.

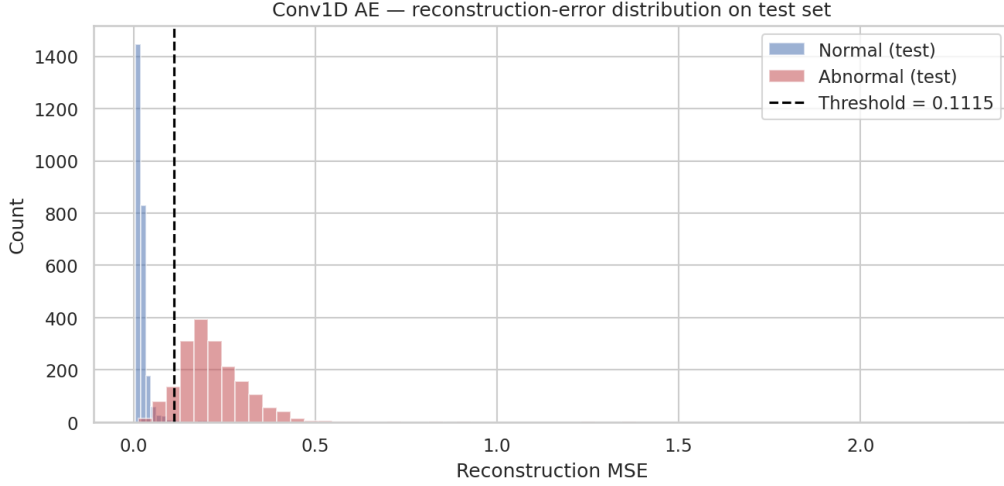


Figure 4: Test-set reconstruction-error distributions (Conv1D AE) with the frozen threshold.

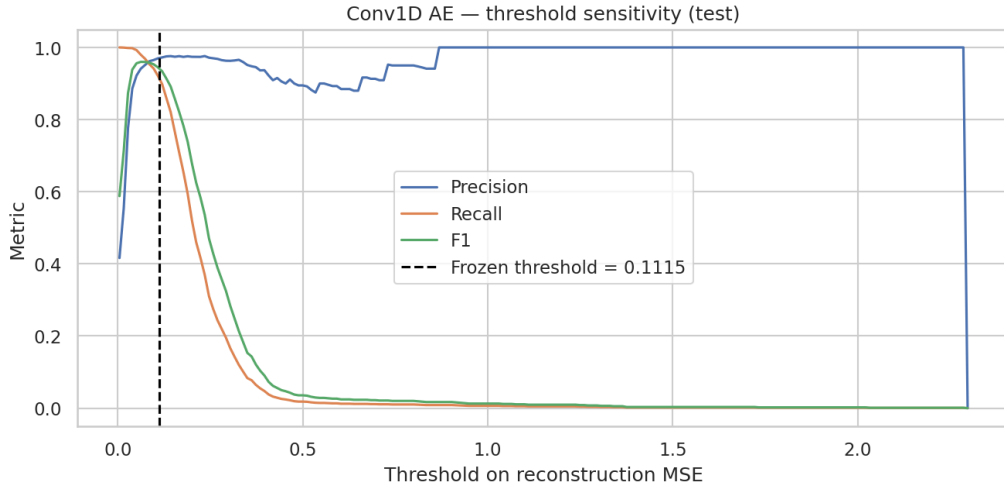


Figure 5: Precision, recall and F1 as a function of threshold on the test set. The dashed line marks the frozen validation-F1 threshold.

8.2 Conv1D vs. LSTM

Table 8 reports the test-set performance of the two autoencoders under matched protocol. Conv1D beats LSTM on every metric while using more parameters (91,729 vs. 14,033) and comparable wall-clock training time on CPU.

Table 8: Conv1D AE vs. LSTM AE on the untouched test set. Threshold selected by validation-F1 argmax for each model.

Model	Precision	Recall	F1	ROC-AUC	PR-AUC	Params	Time (s)
Conv1D AE	0.971	0.916	0.943	0.988	0.967	91,729	11.6
LSTM AE	0.918	0.907	0.912	0.965	0.934	14,033	8.9

Qualitatively the Conv1D AE cleanly reconstructs normal beats and fails on abnormal beats. Figures 6a and 6b show one representative test example of each type.

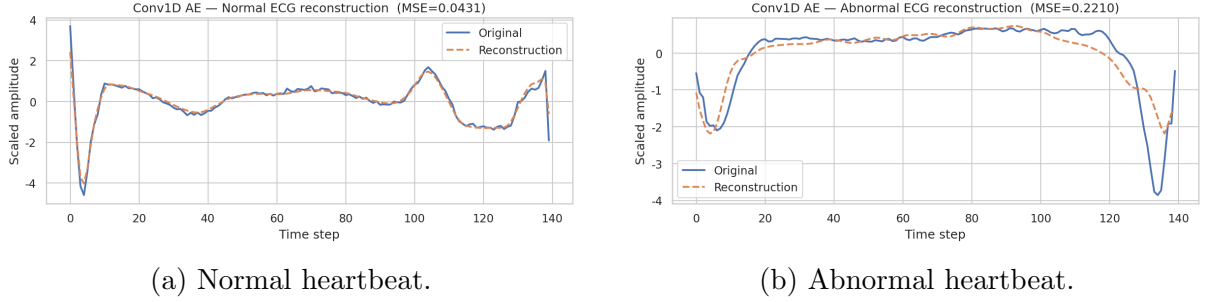


Figure 6: Conv1D AE reconstructions on the test set.

The confusion matrix at the frozen threshold is given in Figure 7: 2 575 TN, 52 FP, 157 FN, 1 716 TP.

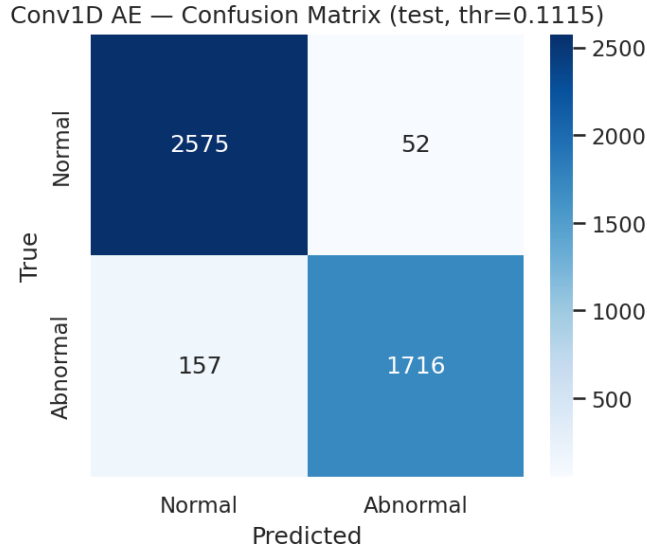


Figure 7: Conv1D AE confusion matrix on the test set.

8.3 Classical baseline comparison

Table 9 lists all four models on the same test set.

Table 9: Classical baseline comparison on the untouched test set.

Method	Precision	Recall	F1	ROC-AUC	PR-AUC
Conv1D AE	0.971	0.916	0.943	0.988	0.967
LSTM AE	0.918	0.907	0.912	0.965	0.934
One-Class SVM	0.938	0.986	0.961	0.988	0.972
Isolation Forest	0.534	0.543	0.538	0.696	0.531

Discussion. OC-SVM is a strong benchmark on ECG5000: on 140-dimensional standardised signals the RBF-boundary approach captures the normal manifold well. It also has the highest recall of the four detectors, but it does so by trading off precision: 123 false positives against 52 for the Conv1D AE. Isolation Forest underperforms substantially, which is consistent with its known weakness on high-dimensional structured signals: its axis-aligned splits struggle to isolate anomalies whose morphology — rather than the value of a single feature — differs from normal. The LSTM AE, though temporally motivated, has far fewer parameters and does not fully match the Conv1D AE despite the sequence structure of ECG.

8.4 Failure-case analysis

We rank Conv1D-AE false negatives by lowest reconstruction error and false positives by highest error to obtain the most informative examples; Figure 8 plots two of each. All 52 false positives are normal beats (class 1) whose morphology contains mild-to-moderate deviations from the typical waveform; the AE faithfully reproduces the low-frequency envelope but not their higher-frequency detail, driving MSE above the threshold.

The 157 false negatives break down by original class as follows.

Table 10: False negatives broken down by original abnormal class (Conv1D AE).

Class	# test rows	# false negatives
2 (R-on-T PVC)	1 590	66
3 (PVC)	86	12
4 (SP/EB, ectopic)	175	72
5 (Unclassifiable)	22	7
Total	1 873	157

Although class 4 accounts for only 9.3% of the abnormal test population, it contributes 46% of the false negatives. This is consistent with the physiological description of class 4: supra-ventricular premature / ectopic beats can retain a near-normal QRS morphology because they originate above the ventricles, so their similarity to normal beats makes them precisely the beats a reconstruction-based detector is least able to catch. We refrain from any clinical claim beyond the boundary of this dataset.

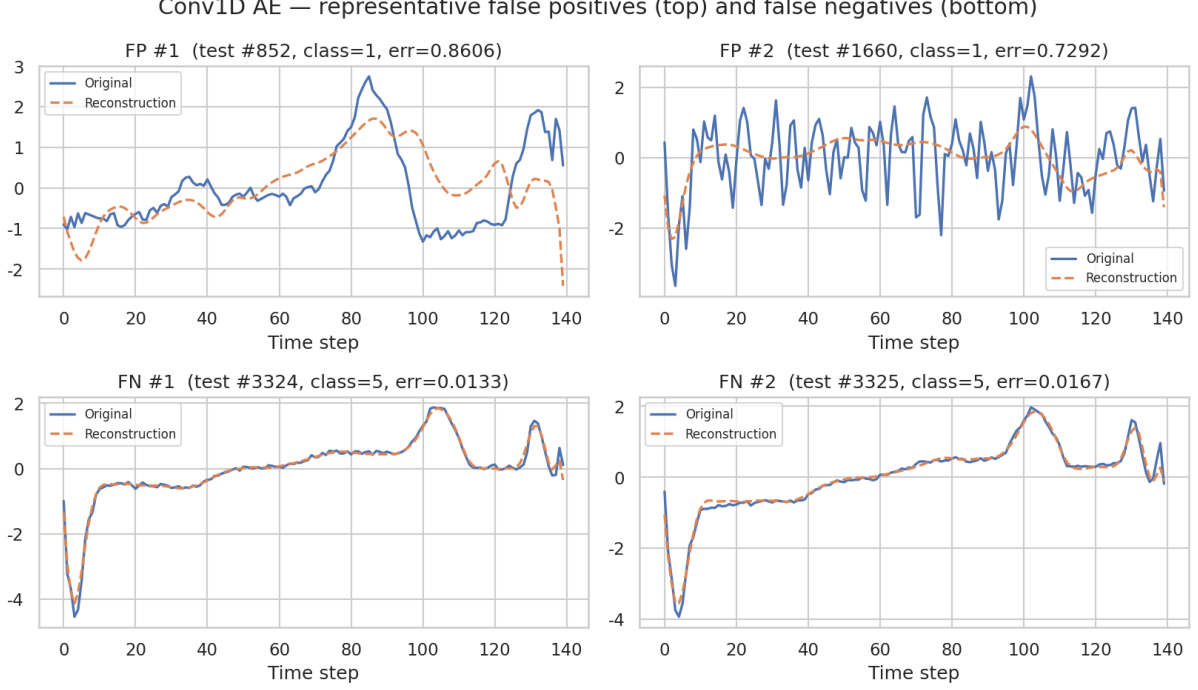


Figure 8: Representative false positives (top) and false negatives (bottom) of the Conv1D AE on the test set. Each panel plots the scaled input against the model reconstruction.

8.5 Capacity ablation

Table 11 sweeps the latent dimension of the Conv1D AE over $\{4, 8, 16, 32, 64\}$; every variant is retrained under a shorter budget (40 epochs, patience 8) and evaluated under the same threshold protocol.

Table 11: Latent-dimension ablation (Conv1D AE). *Sep.* $z = (\mu_{\text{abn}} - \mu_{\text{norm}})/\sigma_{\text{norm}}$ on the validation reconstruction errors.

Latent dim	Params	Sep. z	F1	ROC-AUC	PR-AUC
4	37 957	5.02	0.960	0.986	0.962
8	55 881	5.67	0.936	0.983	0.956
16	91 729	7.65	0.947	0.987	0.963
32	163 425	3.83	0.934	0.974	0.933
64	306 817	5.24	0.947	0.982	0.951

The best F1 in this ablation is attained by the smallest capacity (latent = 4), while the highest separation z-score is at latent = 16. Increasing capacity to 32 or 64 does not monotonically improve detection — consistent with the reconstruction paradox discussed below.

9 The reconstruction paradox

An autoencoder trained *only* on normal data is expected to reconstruct normal signals accurately and abnormal signals poorly; the gap between the two error distributions is what enables detection. When the model becomes too flexible, however, its decoder can learn generic low-level primitives (short piece-wise linear or oscillatory patterns) that happen to reconstruct abnormal beats surprisingly well even without ever training on them. The $s(\text{normal})-s(\text{abnormal})$ gap then shrinks and detection degrades. Our ablation exhibits exactly this effect: the intermediate latent 16 gives the largest normal / abnormal separation, while increasing latent to 32 halves the separation- z from 7.65 to 3.83.

Controllable factors that mitigate the paradox in practice:

- **Smaller latent dimension:** forces the bottleneck to encode only the salient normal morphology.
- **Fewer or narrower convolutional filters:** limits the compositional expressive power of the decoder.
- **Dropout** [15] and weight decay: discourage overfitting the normal training distribution too tightly and reduce spurious generalisation.
- **Early stopping** on *normal-only* validation loss: prevents the network from finding representations that reconstruct *everything* equally well.
- **Clean training data:** contamination by abnormal training beats teaches the decoder to reconstruct anomalies faithfully, which is precisely what we do not want.

The paradox implies that model selection for reconstruction-based anomaly detection must not target reconstruction quality alone; it must target the normal-vs.-abnormal separation of the error distribution.

10 Self-attention discussion

A natural extension is to inject a self-attention block [16] between the last convolutional stage of the encoder and the bottleneck. Attention computes $\text{Att}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d})V$, where Q, K, V are linear projections of the encoded sequence. On an ECG heartbeat one could hope that attention would put high weight on physiologically salient regions (the QRS complex, T-wave onset), producing a latent code sensitive to morphology rather than absolute sample amplitude. Two effects, however, deserve caution:

1. **Additional capacity, additional paradox risk.** Adding attention increases the parameter count and gives the decoder more ways to reconstruct out-of-distribution beats. Our ablation already showed that raw capacity can hurt anomaly separation; attention is a particular way of increasing capacity.
2. **Attention $\neq$ explanation.** It is tempting to read the softmax weights as an explanation of the model’s decision, but attention weights are neither necessary nor sufficient for faithful explanations [8]. Two models that agree on outputs can

produce very different attention maps; ablating the top-attended tokens does not always change the output.

Experimental status. No self-attention experiment was executed in this project. A reasonable future setup would be: (i) add a single-head attention block with 4 heads and head-dim 16 to the encoder; (ii) keep every other hyperparameter identical to the Conv1D AE; (iii) compare detection metrics and normal/abnormal separation z ; (iv) audit whether high-attention time steps correlate with the QRS complex.

11 Limitations

- ECG5000 is a small, curated dataset. All conclusions are stated within its experimental setting only; no clinical implication is claimed.
- The Conv1D-AE flattens the encoder before the bottleneck, which introduces a fully connected layer whose parameter count dominates the model. A fully convolutional bottleneck would be a natural next design.
- The LSTM AE uses a single layer with 32 hidden units; deeper recurrent stacks were not explored.
- Training is CPU-only; GPU experiments would allow larger hyperparameter sweeps.
- Only one random seed is used. A full statistical study would average metrics over multiple seeds and report standard deviations.

12 Conclusion

We built and executed a leakage-free reconstruction-based anomaly detector on ECG5000. Under a matched experimental protocol the Conv1D AE (F1=0.943) beats the LSTM AE (F1=0.912) and Isolation Forest (F1=0.538), and remains competitive with One-Class SVM (F1=0.961). Validation-F1-optimal thresholding produced the strongest validation F1 across four systematically compared strategies and generalised cleanly to the test set. A latent-dimension ablation illustrated the reconstruction paradox: more capacity does not imply better anomaly detection. Error analysis pinpointed abnormal class 4 (supraventricular premature / ectopic beats) as the dominant failure mode — the class morphologically closest to the normal training distribution. Every reported result is directly reproducible from the accompanying notebook.